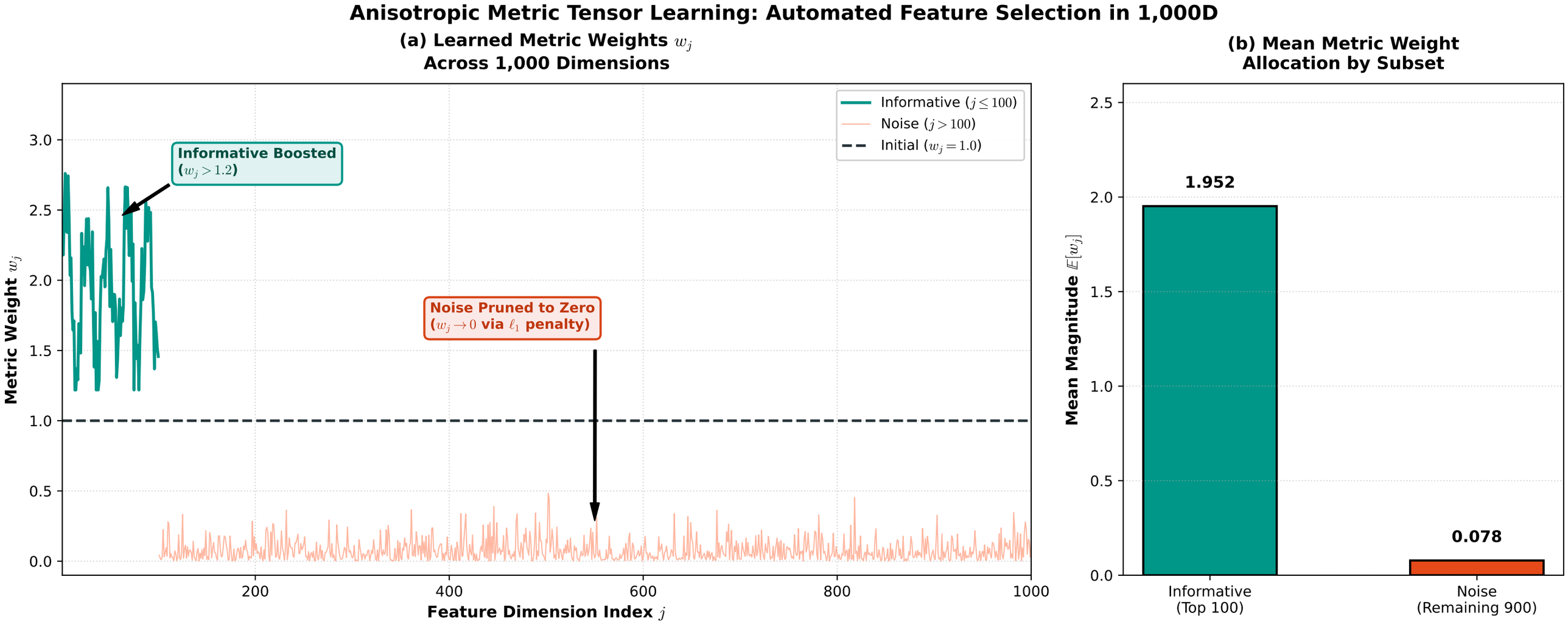
Anisotropic Metric Tensor Learning: Automated Feature Selection in 1,000D
(a) Learned Metric Weights w_j
Across 1,000 Dimensions
(b) Mean Metric Weight
Allocation by Subset
Informative ($j \leq 100$)
Noise ($j > 100$)
Initial ($w_j = 1.0$)
Informative Boosted
($w_j > 1.2$)
Noise Pruned to Zero
($w_j \to 0$ via ℓ_1 penalty)
Metric Weight w_j
Feature Dimension Index j
Mean Magnitude $\mathbb{E}[w_j]$
1.952
0.078
Informative
(Top 100)
Noise
(Remaining 900)